

Special Seminar on Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

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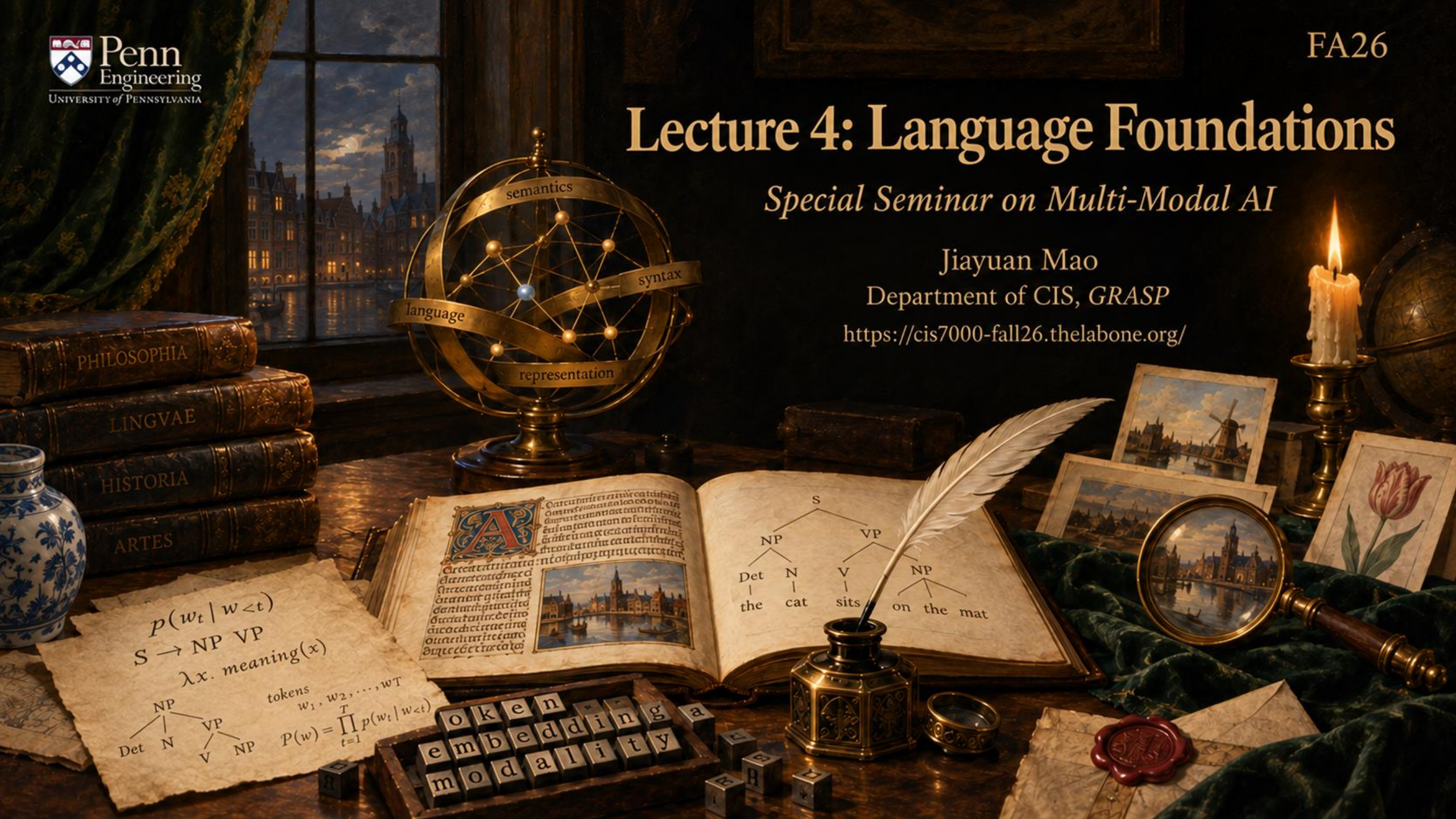
Lecture 4: Language Foundations

Special Seminar on Multi-Modal AI

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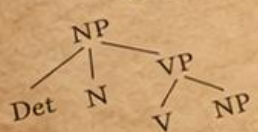
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$$p(w_t | w_{<t})$$

$$S \rightarrow NP VP$$

$$\lambda x. \text{meaning}(x)$$



tokens $w_1, w_2, \dots, w_T$

$$P(w) = \prod_{t=1}^T p(w_t | w_{<t})$$



Enrollment

We have (finally!) finalized the enrollment

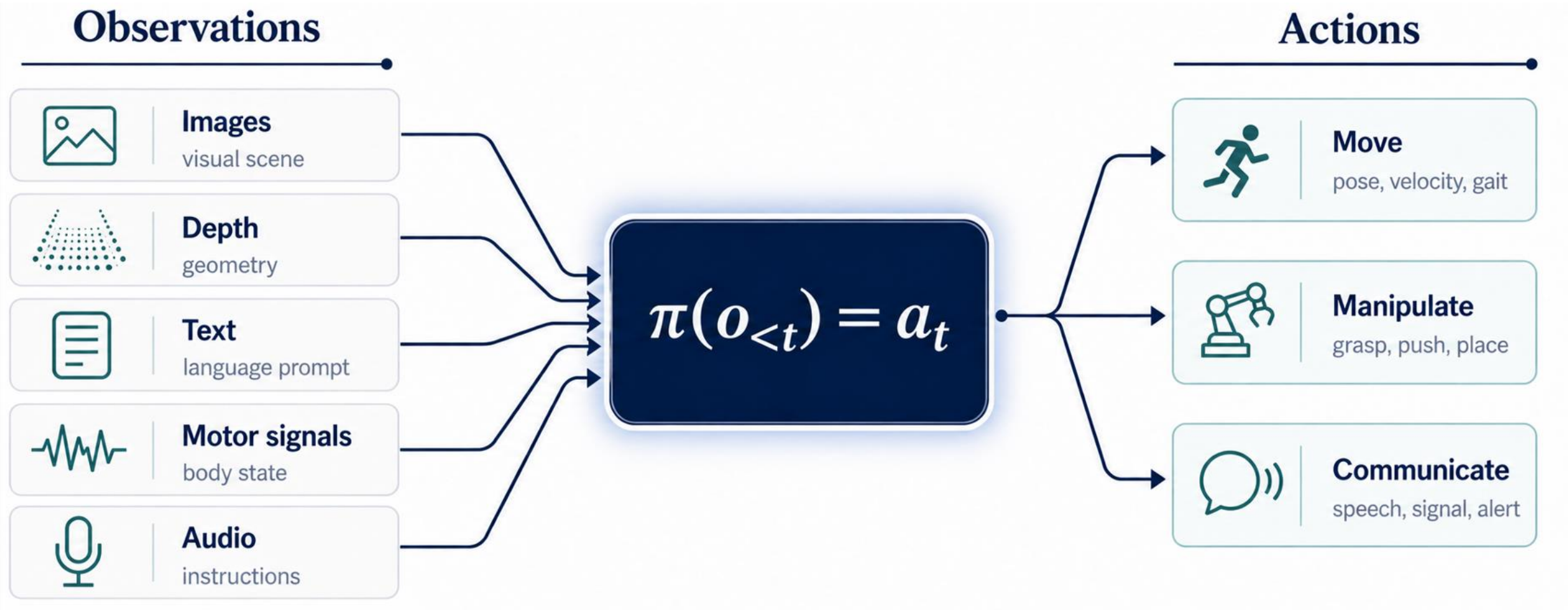
- Reading assignments for Lecture 2, 3 and 4 will not count
- We will be officially grading your reading responses starting next week
- Team up! (Each project should be ≤ 4 students in general)

Goal of This Course

Use Machine Learning to Build Intelligent Agents

Abstract View of An Agent: Multimodal AI

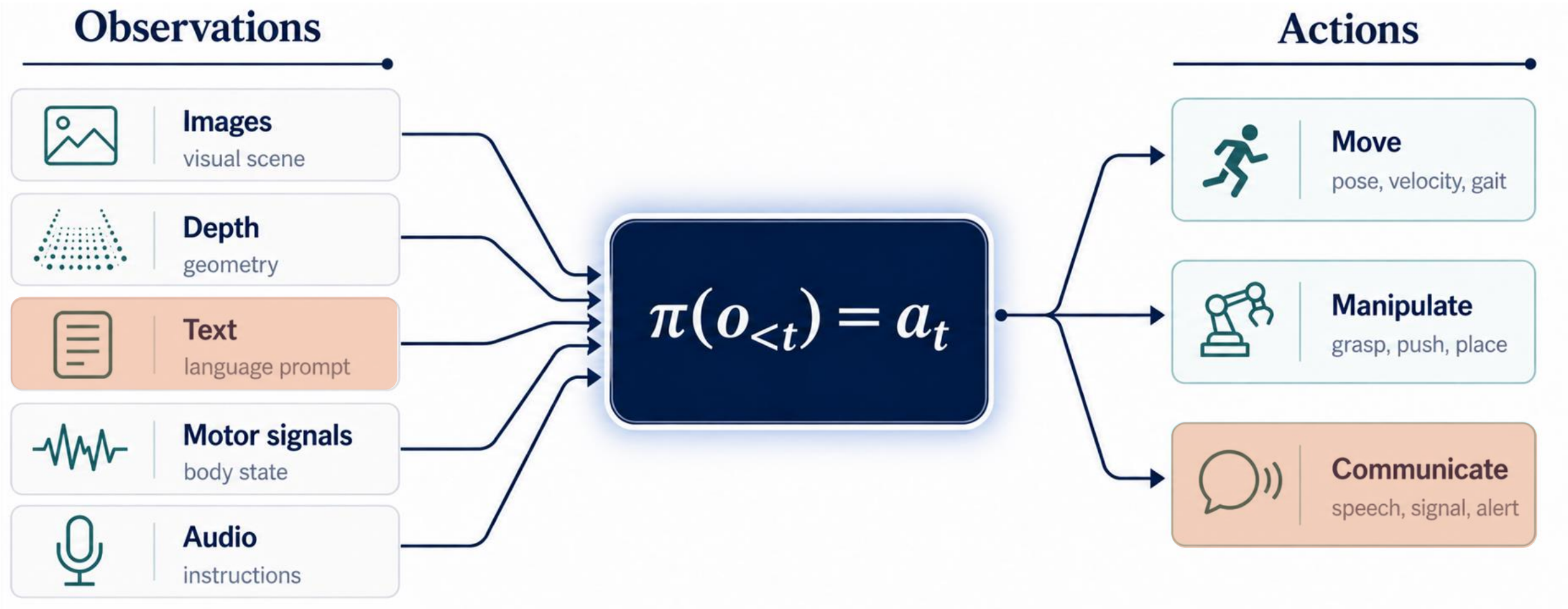
π : observations $\rightarrow$ action



Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Abstract View of An Agent: Multimodal AI

π : observations $\rightarrow$ action



Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Goal of This Lecture

Use Machine Learning to Model Language

Today's Agenda

Part 1: Principles for (Language) Tokenization: BPE

Part 2: Principles for Language Modeling

- The attention mechanism and its caveats

Part 3: Principles for Large Language Models

Today's Agenda

Part 1: Principles for (Language) Tokenization: BPE

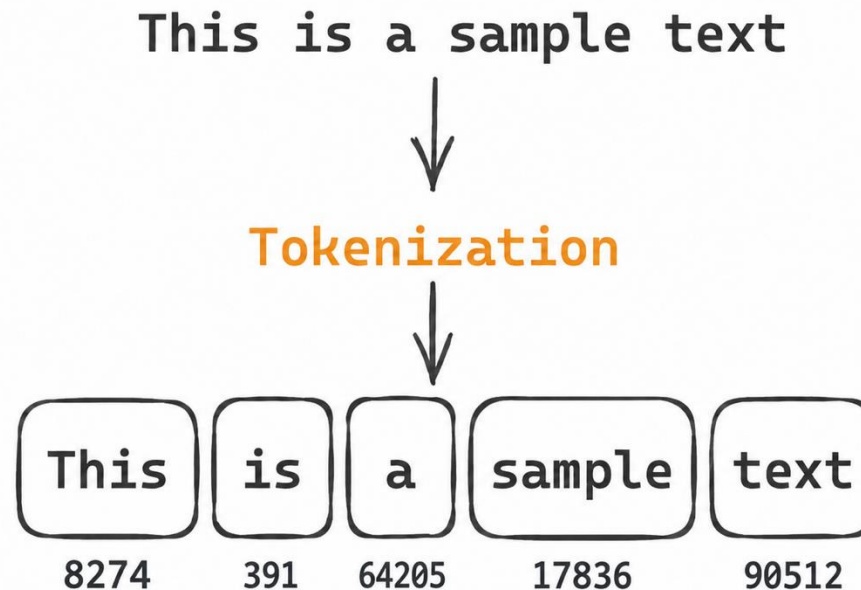
Part 2: Principles for Language Modeling

- The attention mechanism and its caveats

Part 3: Principles for Large Language Models

Tokenization

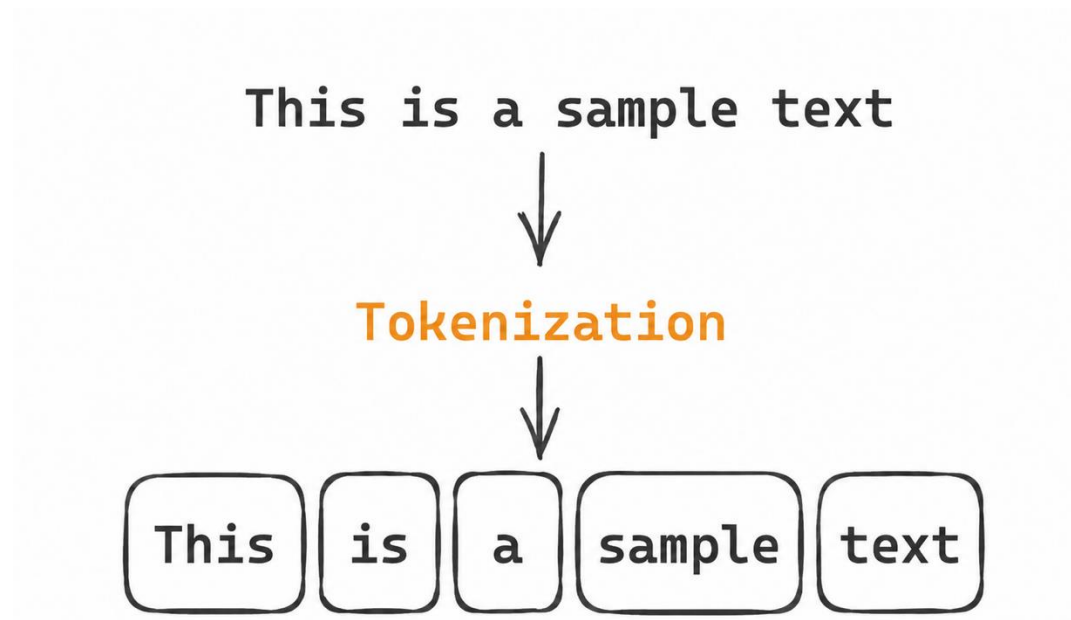
- We need to represent texts as discrete indices
- The simplest idea:
`[word2index[w] for w in text.split()]`



Tokenization

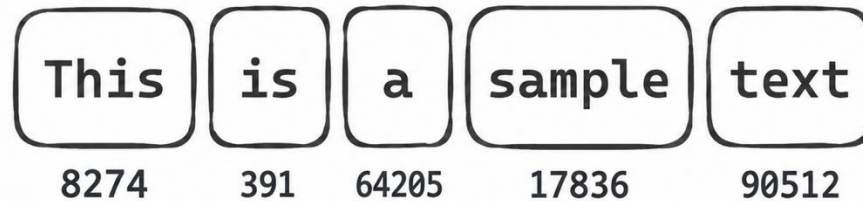
- We need to represent texts as a discrete indices
- The simplest idea:

```
[word2index[w] for w in text.split()]
```



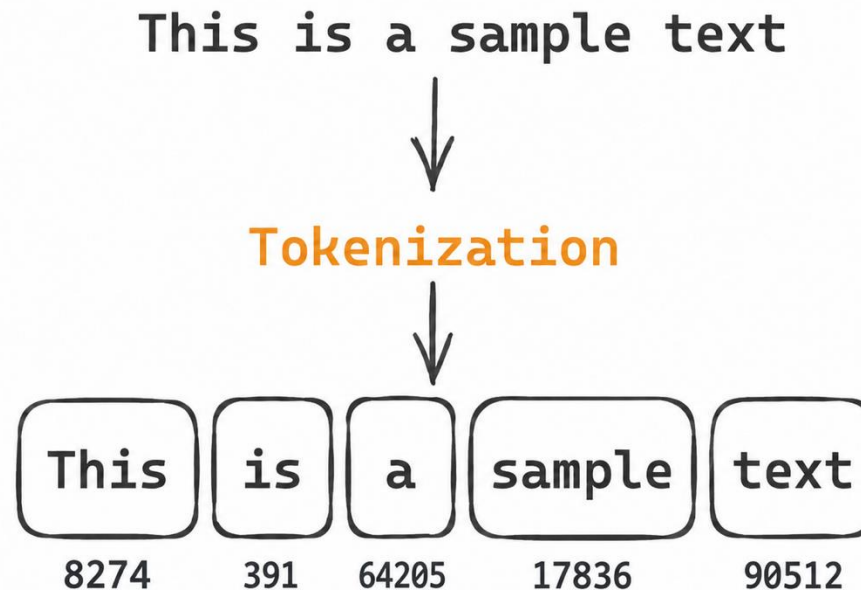
Tokenization

- We need to represent texts as a discrete indices
- The simplest idea:
[`word2index[w]` for `w` in `text.split()`]



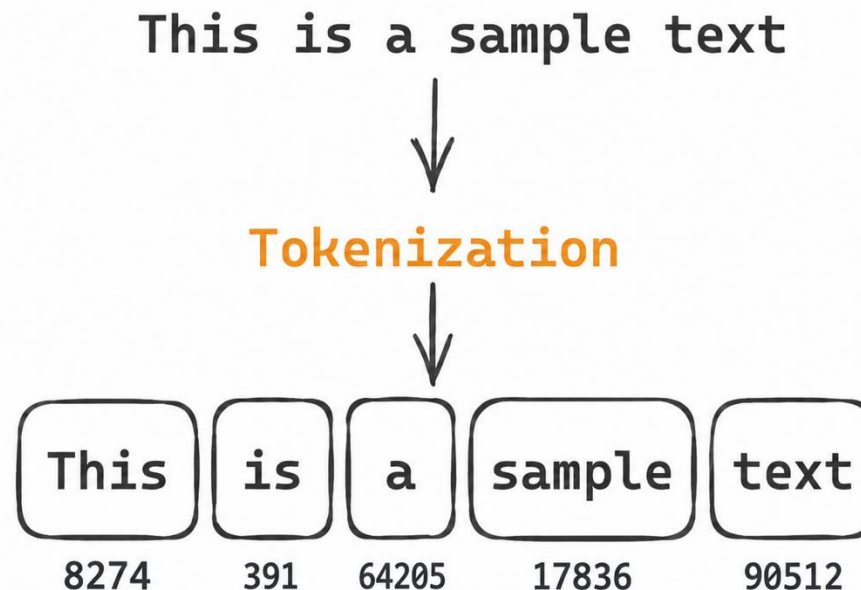
Tokenization

- We need to represent texts as a discrete indices
- The simplest idea:
`[word2index[w] for w in text.split()]`



Tokenization

- We need to represent texts as a discrete indices
- The simplest idea:
`[word2index[w] for w in text.split()]`
- How about different languages? How about punctuations?



Byte-Pair Encoding

- Previous approach: human-defined rules to handle punctuations, hyphenation, numbers, special characters, and *rare words*
- BPE Solution: Just start from the basic vocab (e.g., all ASCII), and use the greedy algorithm to construct a vocabulary

The Entropy Intuition behind Byte-Pair Encoding

- BPE is actually a general “compression algorithm”
- BPE makes frequent strings cheaper to decode
- The model “sees” shorter sequences and denser token statistics

Tokenization and the Noiseless Channel

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Mrinmaya Sachan^E Ryan Cotterell^E

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The Entropy Intuition behind

Tokenization and the Noiseless Channel

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- The “efficiency” of the tokenization process
 - Intuitively, what’s the entropy after rewriting the inputs with the vocabulary
- is correlated with the task performance

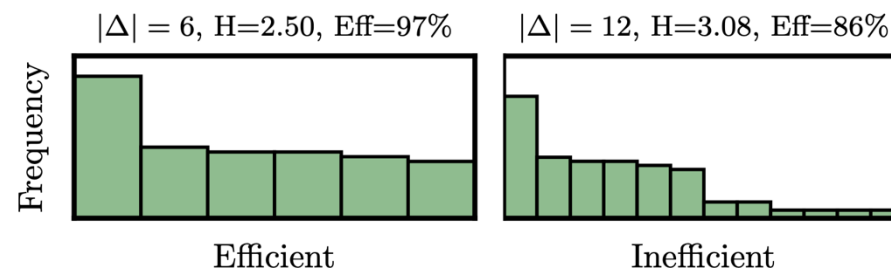


Figure 1: Examples of unigram distributions with efficient and inefficient channel usage.

The Entropy Intuition behind

Tokenization and the Noiseless Channel

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- The “efficiency” of the tokenization process
 - Intuitively, what’s the entropy after rewriting the inputs with the vocabulary
- is correlated with the task performance
- Note: This is not generally true for arbitrary modality data

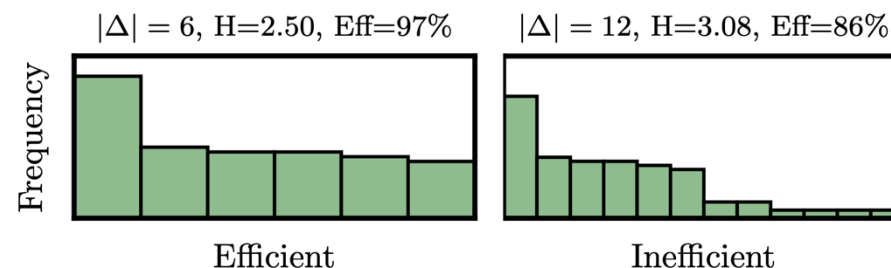


Figure 1: Examples of unigram distributions with efficient and inefficient channel usage.

Byte-Level BPE in Modern LLMs

- To model every possible Unicode code (e.g., emojis) directly, the base vocabulary would be very large
- Solution: Start from 256 UTF-8 byte values, then learn BPE merges on byte sequences
- Used by GPT-2

Fairness in BPE

- Since the construction is statistical, it will assign less vocabulary for underrepresented languages
- For example, handling of space is different
- Cross-Lingual fairness is an active research area

Parity-Aware Byte-Pair Encoding: Improving Cross-lingual Fairness in Tokenization

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Joel Niklaus² **Sina Ahmadi³** **Antoine Bosselut^{1†}** **Rico Sennrich^{3†}**

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Principles of Language Tokenization

- Improved from human-defined language-specific heuristics to general algorithms: byte-level BPE
 - However! Still based on human-defined text encodings (e.g., UTF-8.txt)
 - Extension to arbitrary binary files is still unclear
- Compression and removing easy-to-predict learning objectives is helpful

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Part 1: Principles for (Language) Tokenization: BPE

Part 2: Principles for Language Modeling

- The attention mechanism and its caveats

Part 3: Principles for Large Language Models

The Attention Mechanism

- Attention has a long history of applications in CV/NLP/Speech/Robotics
- This is a famous paper of applying attention+RNN for NLP

Published as a conference paper at ICLR 2015

NEURAL MACHINE TRANSLATION BY JOINTLY LEARNING TO ALIGN AND TRANSLATE

Dzmitry Bahdanau
Jacobs University Bremen, Germany

KyungHyun Cho **Yoshua Bengio***
Université de Montréal

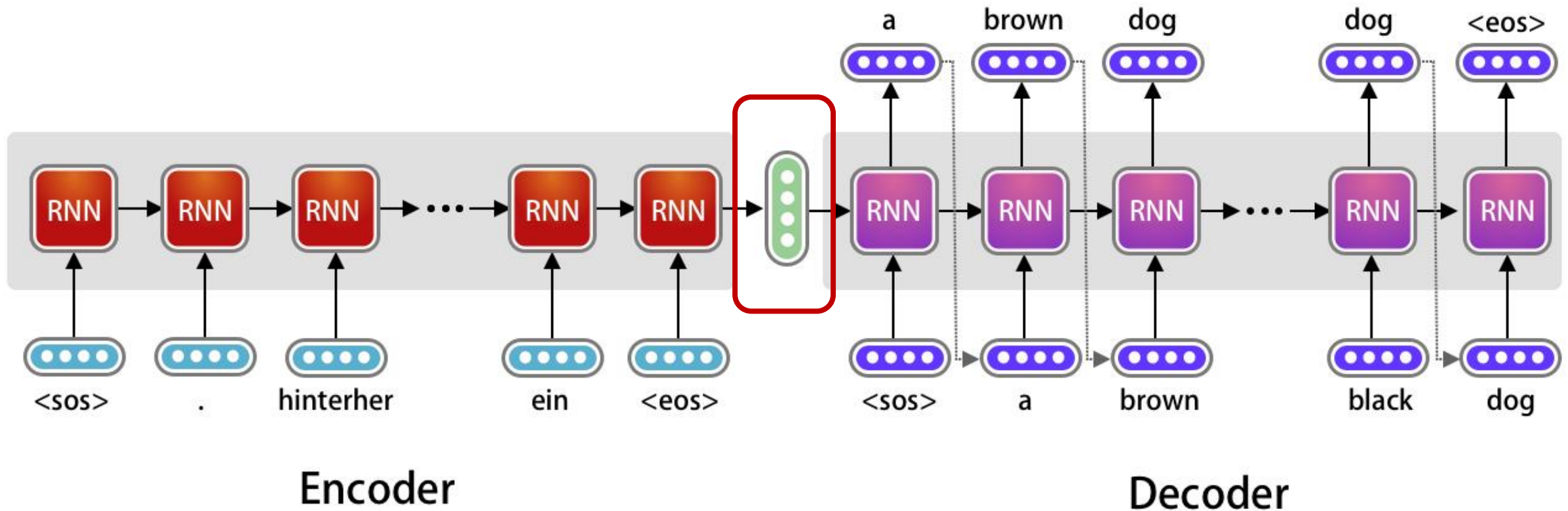
The Attention Mechanism

- Motivation: Sequence-to-Sequence models
- The middle state becomes a “bottleneck”

NEURAL MACHINE TRANSLATION
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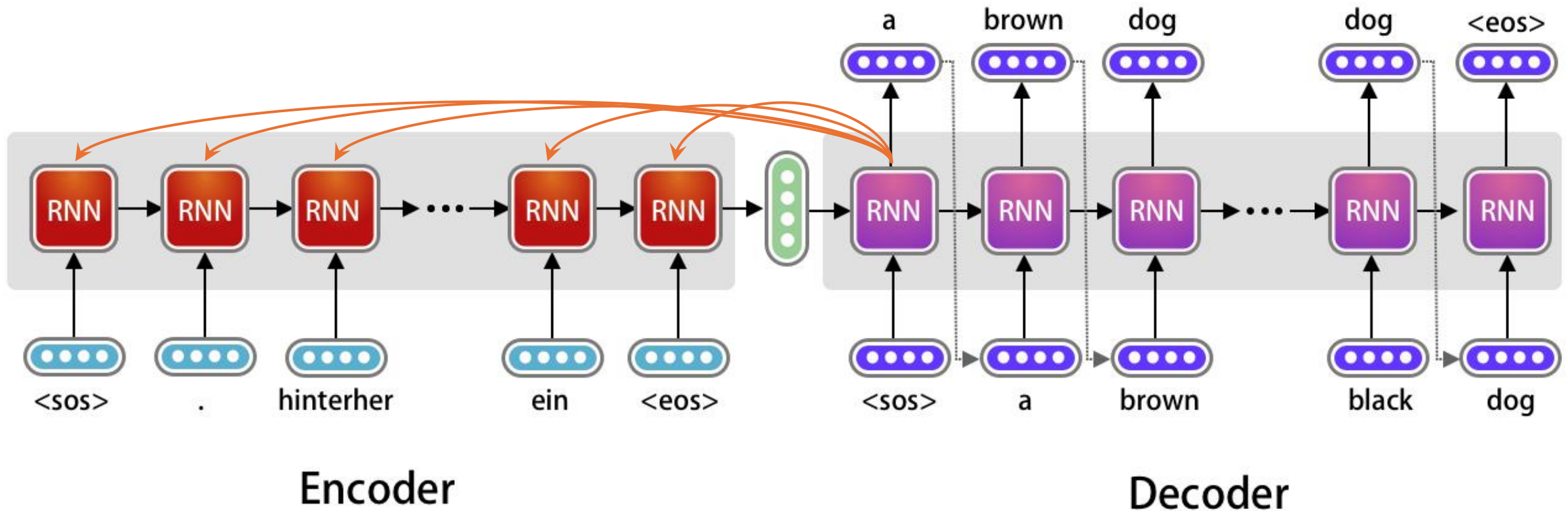
The Attention Mechanism

- Intuition: Align the words in translation

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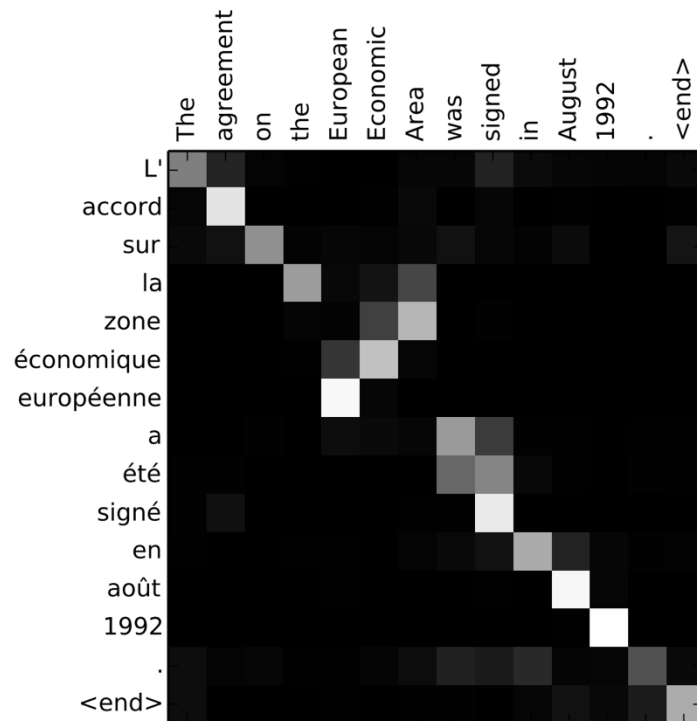
The Attention Mechanism

- Intuition: Align the words in translation
- It automatically discovered the “word alignment”

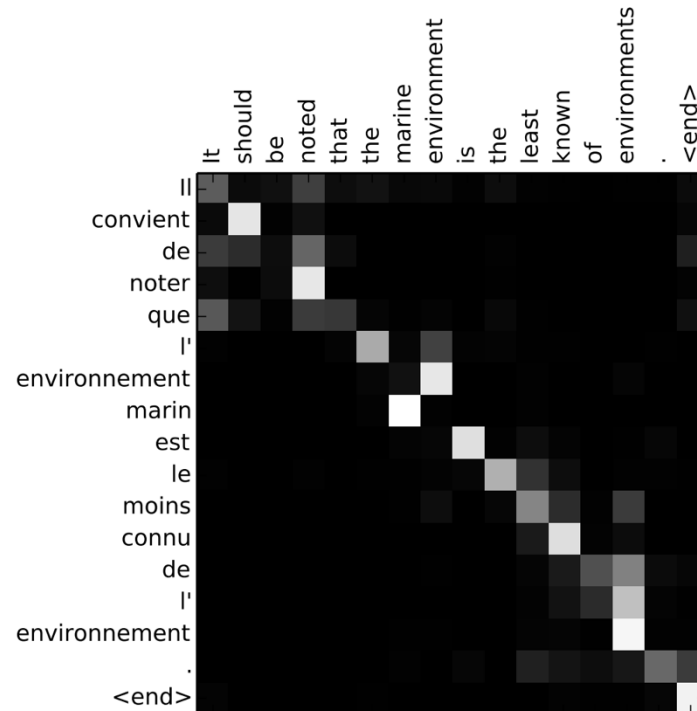
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(a)



(b)

Remove the RNN: Attention is All You Need

Attention Is All You Need

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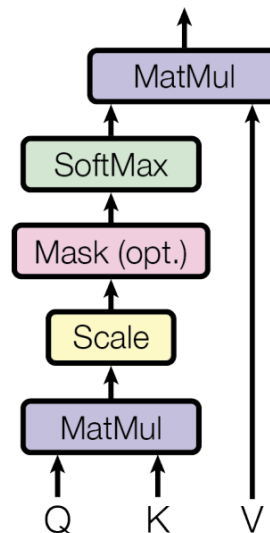
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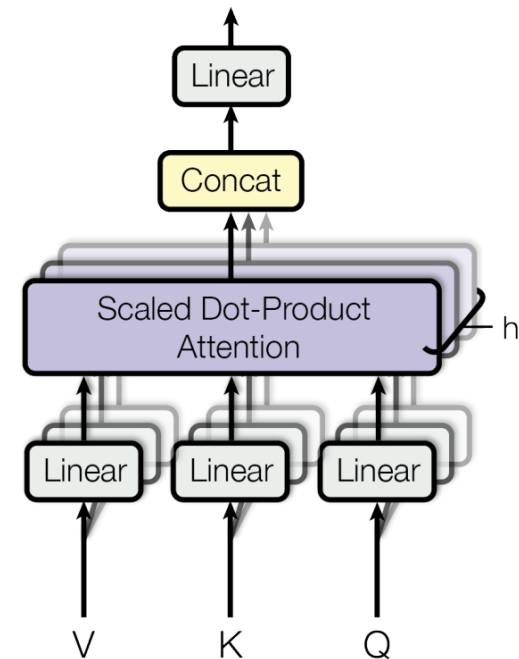
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Scaled Dot-Product Attention



Multi-Head Attention



Remove the RNN: Attention is All You Need

Attention Is All You Need

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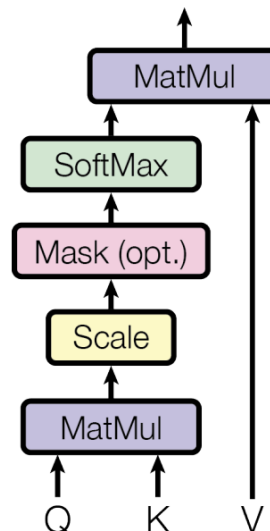
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Scaled Dot-Product Attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Why $\sqrt{d_k}$? For d IID Gaussian variables $x_i \sim N(0, 1)$, $\sum x_i \sim N(0, \sqrt{d})$.
- Since we use Gaussian to initialize neural network weights, we usually assume the activations follow Gaussian distributions too
- Very common assumption / trick!

The Attention Layer May Not Attend to What You Expected

What does BERT look at? An Analysis of BERT's Attention

Kevin Clark[†] Urvashi Khandelwal[†] Omer Levy[‡] Christopher D. Manning[†]

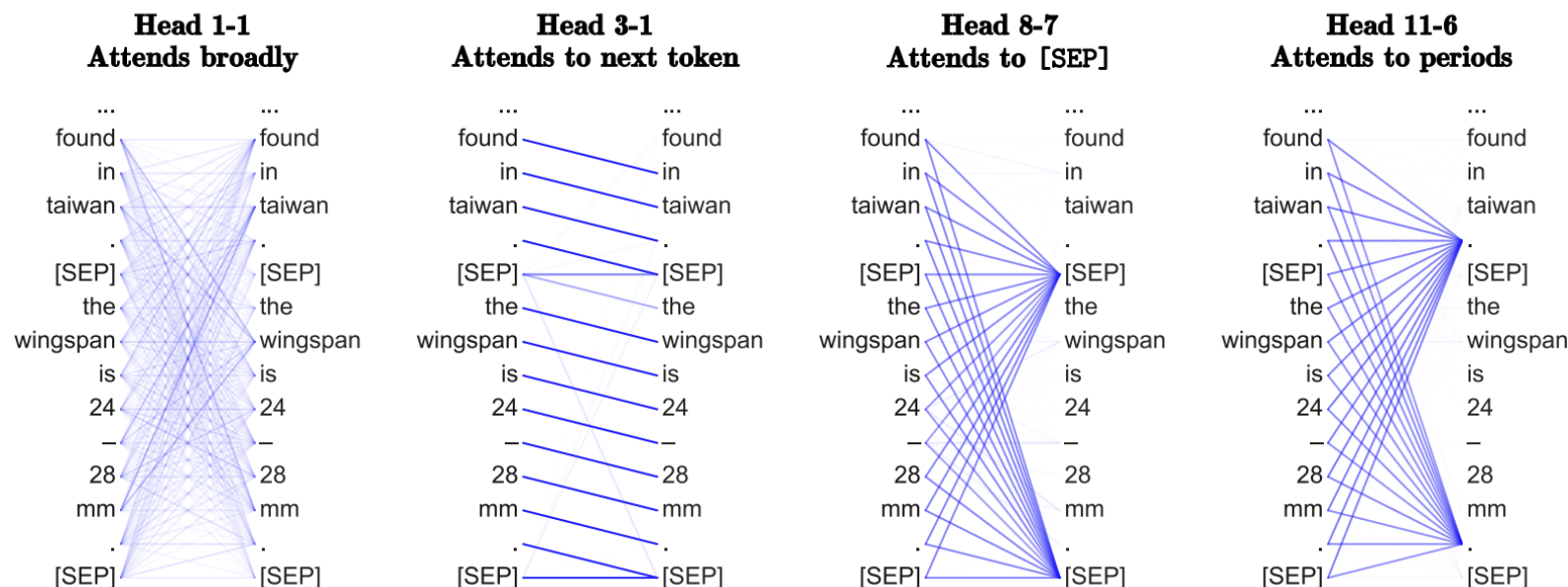
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The Attention Layer May Not Attend to What You Expected



- In general contexts, the attention may not be interpretable
- True for multi-modal models too: e.g., image-text attention

And... Many Attention Heads May Not Be Useful at All

Are Sixteen Heads Really Better than One?

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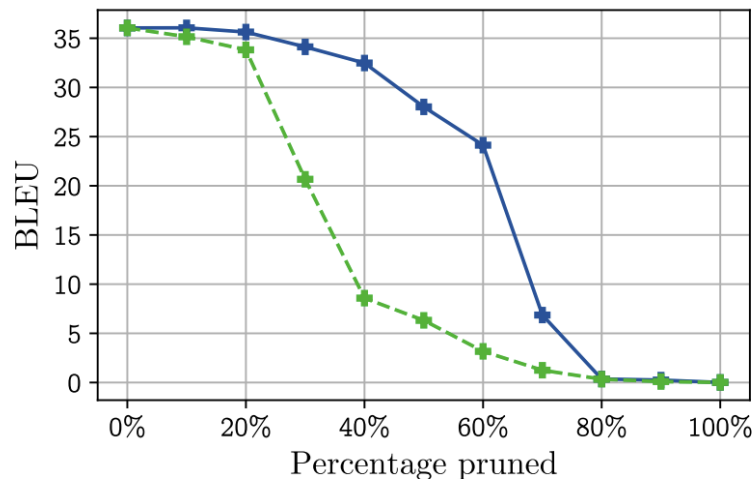
And... Many Attention Heads May Not Be Useful at All

Are Sixteen Heads Really Better than One?

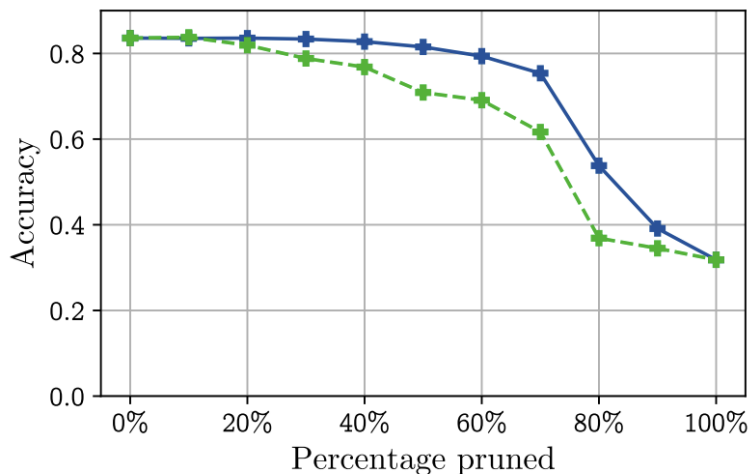
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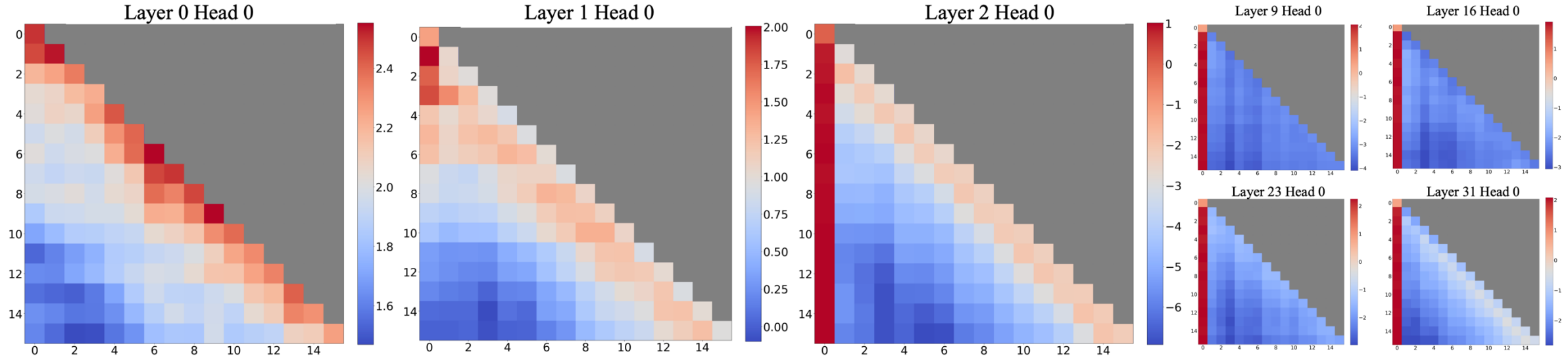
(a) Evolution of BLEU score on newstest2013 when heads are pruned from WMT.



(b) Evolution of accuracy on the MultiNLI-matched validation set when heads are pruned from BERT.

- Removing ~40% of the attention heads do not hurt the performance
- But this might be important for optimization purposes
- Intuition: if we have only 1 head, it must learn (through SGD) a “good” attention; if we have more, we are good as long as one of them works

Attention Sink: Attention in Autoregressive Models



- Attention is heavily biased towards “local” and “first N” tokens
- We can leverage this to *mechanically* restrict the attention window to speed up inference

Principles of Language Modeling: Attention

- Originated from the “word alignment” problem
- Became a general computational building block
- Attention may not work as you (may have) expected

Today's Agenda

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Part 2: Principles for Language Modeling

- The attention mechanism and its caveats

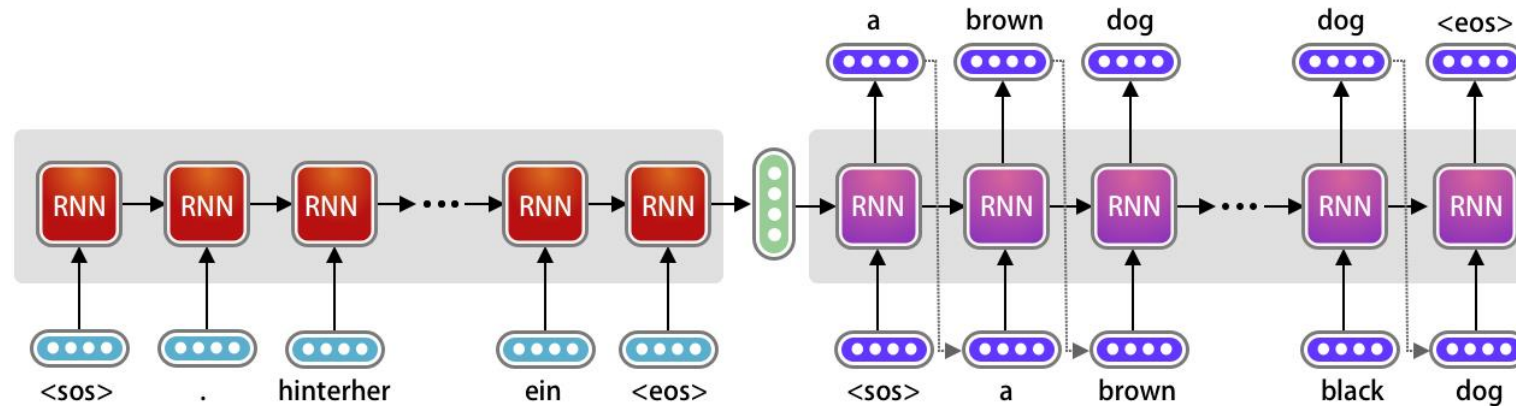
Part 3: Principles for Large Language Models

The Evolution of LLMs

- GPT-2: Language Models are Unsupervised Multitask Learners
- GPT-3: Language Models are Few-Shot Learners
- GPT-4+: With thinking, with tool use, etc.
- Today we will cover some important ideas

From Encoder-Decoder and Decoder-Only LLMs

- The classical ML pipeline: model = feature extractor + predictor



- GPT-2: at the end, we just need a model that can “complete” a sequence

Language Models are Unsupervised Multitask Learners

Alec Radford^{*1} Jeffrey Wu^{*1} Rewon Child¹ David Luan¹ Dario Amodei^{**1} Ilya Sutskever^{**1}

From Encoder-Decoder and Decoder-Only LLMs

- The classical ML pipeline: model = feature extractor + predictor
- GPT-2: at the end, we just need a model that can “complete” a sequence
- And there are naturally occurring data!

Language Models are Unsupervised Multitask Learners

Alec Radford ^{*1} Jeffrey Wu ^{*1} Rewon Child ¹ David Luan ¹ Dario Amodei ^{**1} Ilya Sutskever ^{**1}

”I’m not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbécile** [I’m not a fool].

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: **”Mentez mentez, il en restera toujours quelque chose,”** which translates as, **”Lie lie and something will always remain.”**

“I hate the word ‘**perfume**,’” Burr says. ‘It’s somewhat better in French: ‘**parfum**.’

If listened carefully at 29:55, a conversation can be heard between two guys in French: **“-Comment on fait pour aller de l’autre côté? -Quel autre côté?”**, which means **“- How do you get to the other side? - What side?”**.

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

“Brevet Sans Garantie Du Gouvernement”, translated to English: **“Patented without government warranty”**.

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

From Encoder-Decoder and Decoder-Only LLMs

- The classical ML pipeline: model = feature extractor + predictor
- GPT-2: at the end, we just need a model that can “complete” a sequence
- And there are naturally occurring data!
- The performance were not good at the time, but we saw “scale laws”

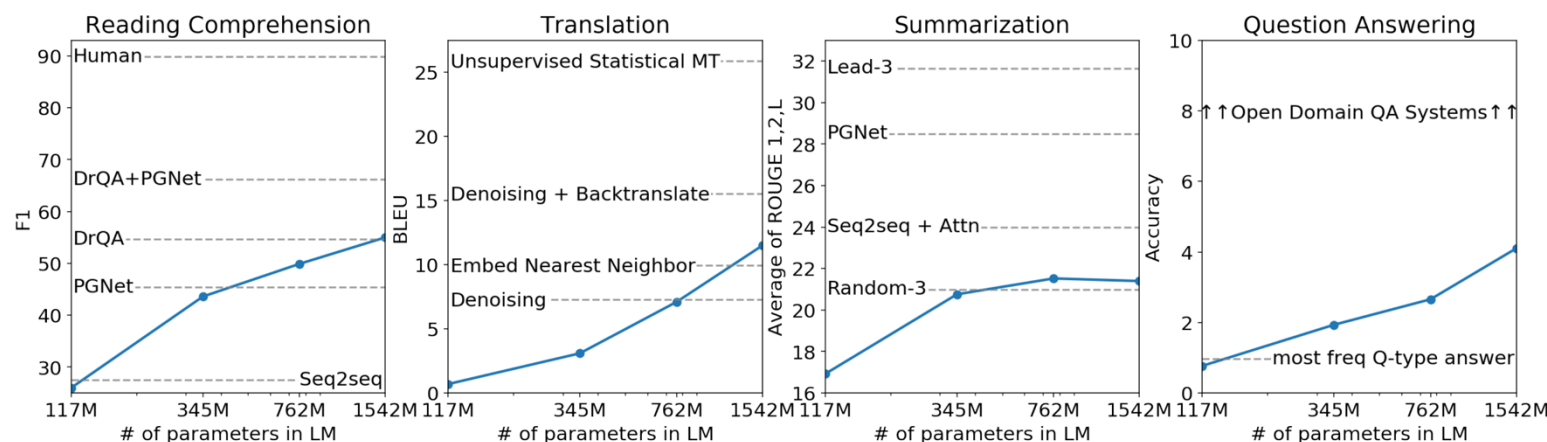


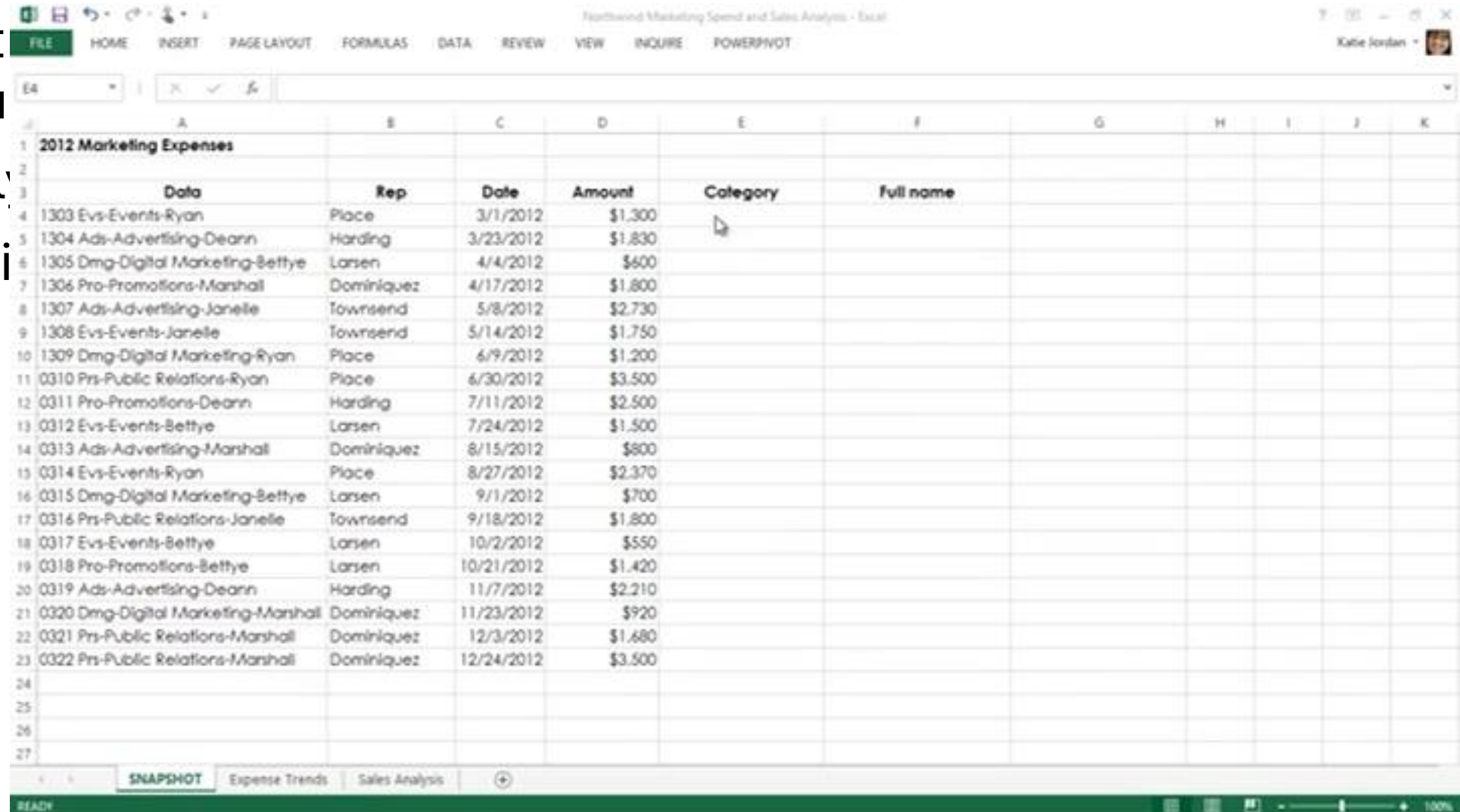
Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.

From Task Prompt to In-Context Learning

- Then we have a new problem: how to set the prefix if we want the LLM to finish a particular task?
- This is generally very hard (for all domains): the specification problem!
- One idea: add input-output examples

From Task Prompt to In-Context Learning

- Then we have to finish a particular task
- This is generally a one-time task
- One idea: add in context



Northwind Marketing Spend and Sales Analysis - Excel

Katie Jordan

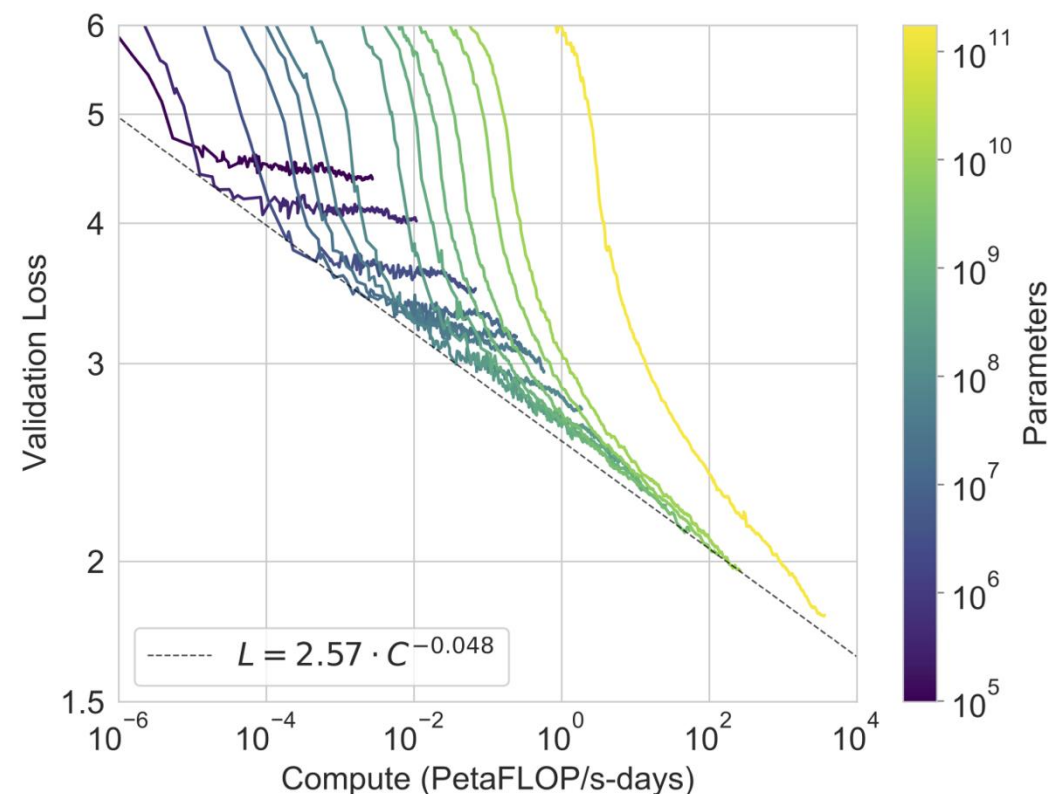
	A	B	C	D	E	F	G	H	I	J	K
1	2012 Marketing Expenses										
2											
3	Data	Rep	Date	Amount	Category	Full name					
4	1303 Evs-Events-Ryan	Place	3/1/2012	\$1,300							
5	1304 Ads-Advertising-Deann	Harding	3/23/2012	\$1,830							
6	1305 Dmg-Digital Marketing-Bettye	Larsen	4/4/2012	\$600							
7	1306 Pro-Promotions-Marshall	Dominiquez	4/17/2012	\$1,800							
8	1307 Ads-Advertising-Janelle	Townsend	5/8/2012	\$2,730							
9	1308 Evs-Events-Janelle	Townsend	5/14/2012	\$1,750							
10	1309 Dmg-Digital Marketing-Ryan	Place	6/9/2012	\$1,200							
11	0310 Prs-Public Relations-Ryan	Place	6/30/2012	\$3,500							
12	0311 Pro-Promotions-Deann	Harding	7/11/2012	\$2,500							
13	0312 Evs-Events-Bettye	Larsen	7/24/2012	\$1,500							
14	0313 Ads-Advertising-Marshall	Dominiquez	8/15/2012	\$800							
15	0314 Evs-Events-Ryan	Place	8/27/2012	\$2,370							
16	0315 Dmg-Digital Marketing-Bettye	Larsen	9/1/2012	\$700							
17	0316 Prs-Public Relations-Janelle	Townsend	9/18/2012	\$1,800							
18	0317 Evs-Events-Bettye	Larsen	10/2/2012	\$550							
19	0318 Pro-Promotions-Bettye	Larsen	10/21/2012	\$1,420							
20	0319 Ads-Advertising-Deann	Harding	11/7/2012	\$2,210							
21	0320 Dmg-Digital Marketing-Marshall	Dominiquez	11/23/2012	\$920							
22	0321 Prs-Public Relations-Marshall	Dominiquez	12/3/2012	\$1,680							
23	0322 Prs-Public Relations-Marshall	Dominiquez	12/24/2012	\$3,500							
24											
25											
26											
27											

SNAPSHOT Expense Trends Sales Analysis

READY

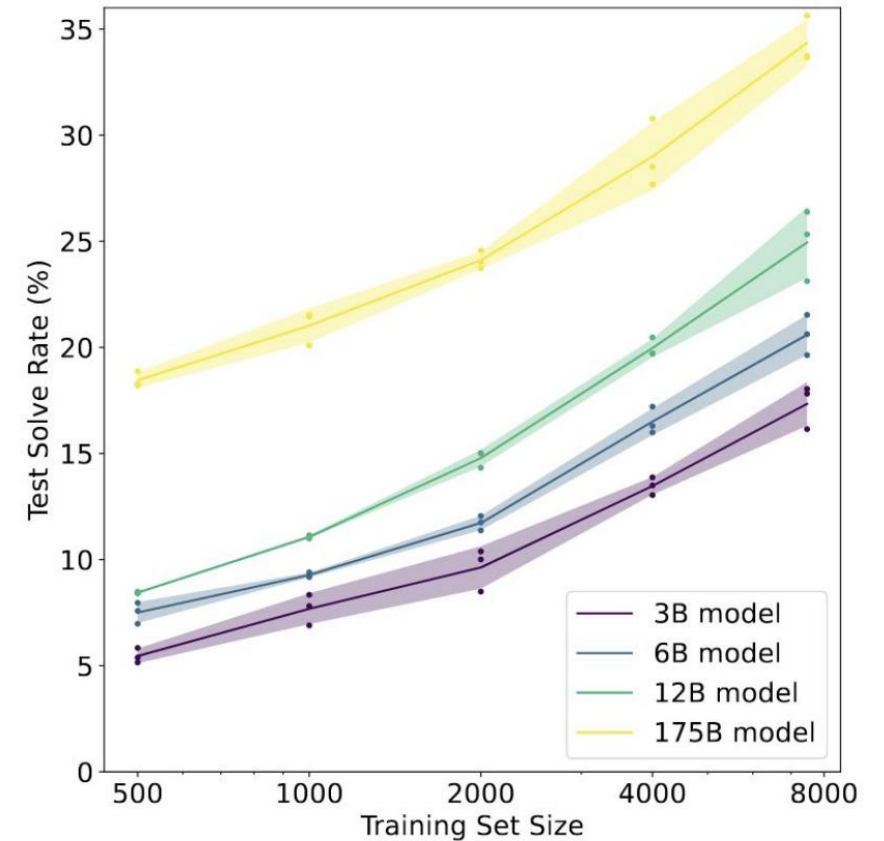
From Task Prompt to In-Context Learning

- Another important contribution: the scaling law
- As we scale up the parameter size and compute (accordingly!), we can get lower validation loss



Chain-of-Thought Reasoning: Motivation

- Fine-tune GPT-3 on GSM8K (arithmetic): (Cobbe et al. 2021)
- Conjecture: to achieve $> 80\%$, needs 100 times more fine-tuning data for 175B model
- Scaling up language model size does not efficiently achieve high performances for certain tasks



Chain-of-Thought Reasoning: Motivation

Chain of Thought Prompting Elicits Reasoning in Large Language Models

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Large Language Models are Zero-Shot Reasoners

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Yutaka Matsuo
The University of Tokyo

Yusuke Iwasawa
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Chain-of-Thought Reasoning: Examples

Examples

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(b) Few-shot-CoT (Wei et al., 2022)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(d) Zero-shot-CoT (KoJima et al., 2022)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

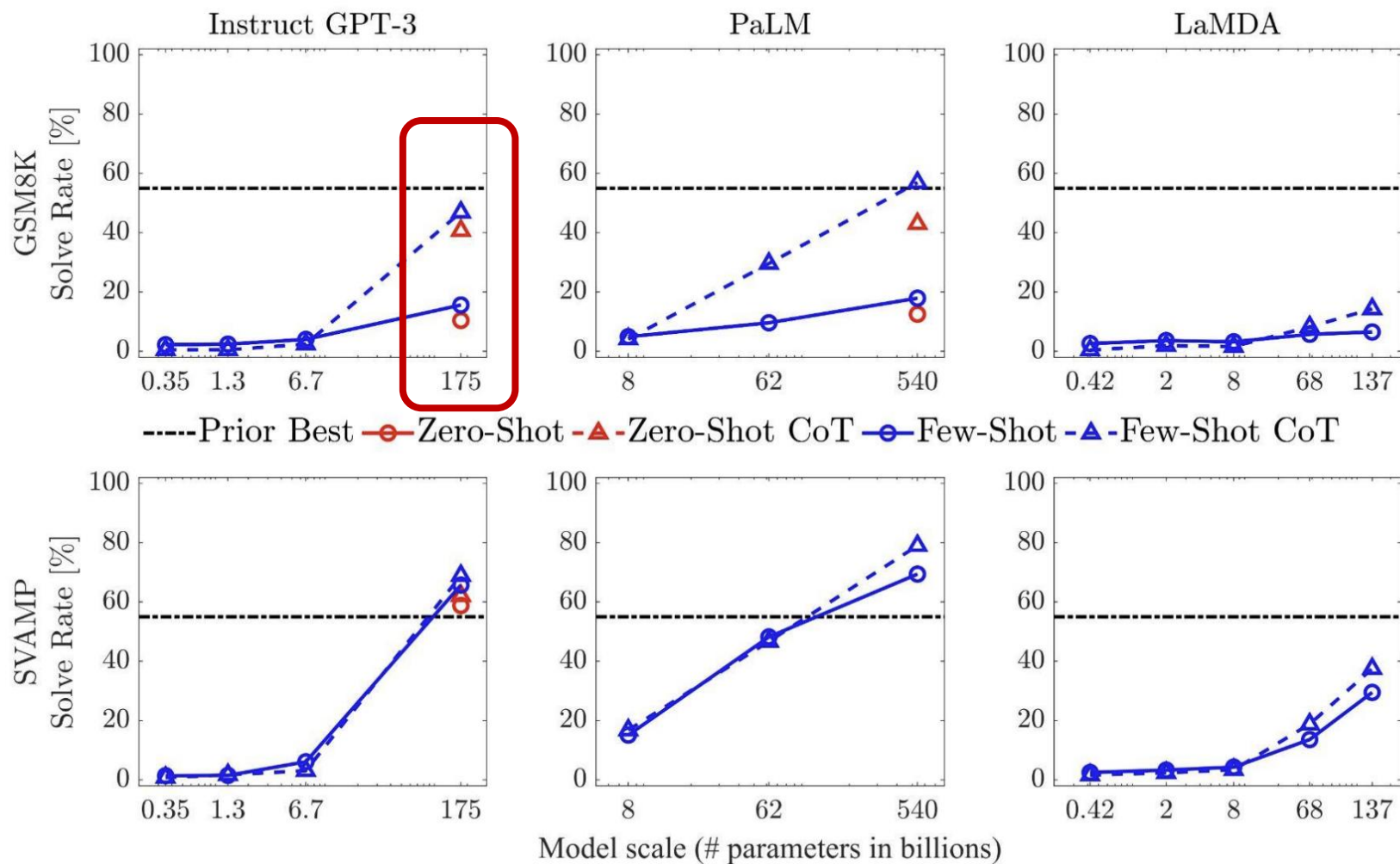
(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

CoT Examples

Step-by-step Answer

Two-stage Prompting Step-by-step Answer

Chain-of-Thought Reasoning: Results



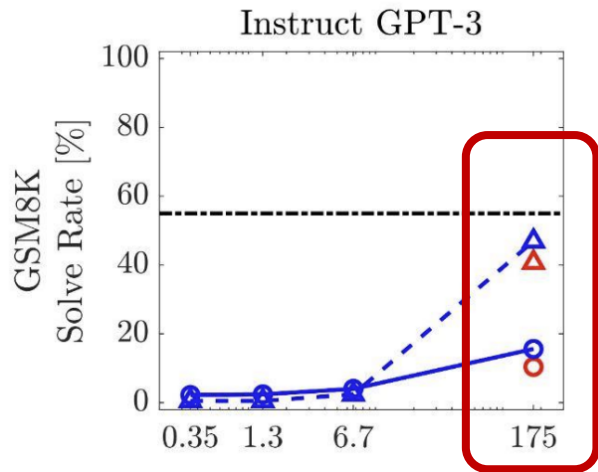
GSM8K

Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

SVAMP

Each pack of dvds costs 76 dollars. If there is a discount of 25 dollars on each pack. How much do you have to pay to buy each pack?

Sometimes Called the “Emergence” or “Grokking”



- Grokking: a “capability” emerges only when the model size is bigger than a threshold
- Many papers tried to explain this, for example:

Are Emergent Abilities of Large Language Models a Mirage?

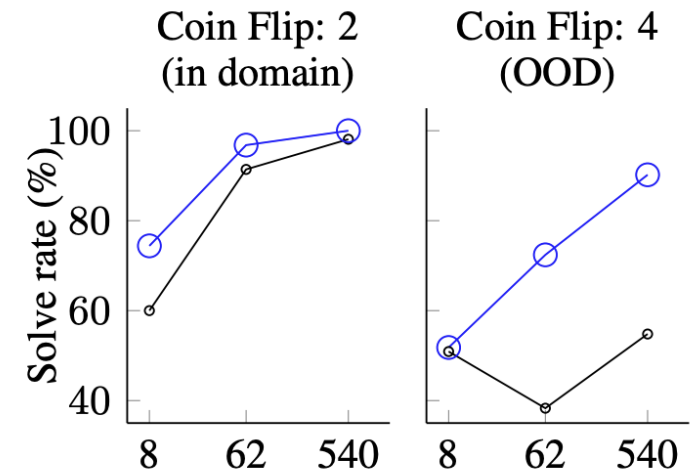
Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo

Computer Science, Stanford University

- It’s a metric-misalignment problem!
- Training: $\sum_i \log p(w_i)$
- Eval: $\min_i accuracy(w_i)$

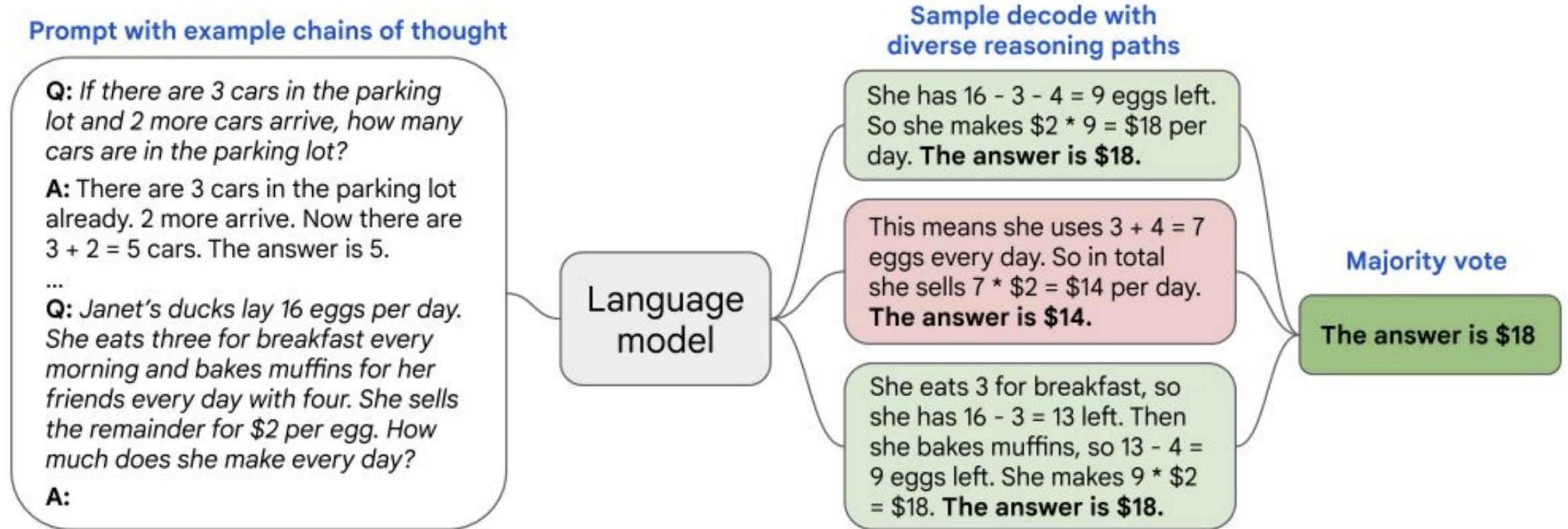
Chain-of-Thought Reasoning: “Out-Of-Domain”

- Motivating Task: “A coin is heads up. Phoebe flips the coin. Osvaldo does not flip the coin. Is the coin still heads up?”
- In-context examples: at most two steps
- Test: 4 steps
- Intuition: leverage the locality of reasoning steps
- Each “primitive” (counting, even number classification) might have been learned by the model



A: The coin was flipped by Phoebe and Osvaldo. So the coin was flipped **2 times**, which is an **even number**. The coin started heads up, so after an even number of flips, it will still be heads up. So the answer is yes.

From CoT to CoT Consistency



- Another important idea in AI/ML, in code generation, in 3D, etc.
- This partially motivated the “multi-agent” debate approach as well

Principles of Large Language Modeling

- GPT-2: Unify the task interface, and leverage naturally occurring data
- GPT-3: Use example-based interface
- GPT-4+: Use thinking to decompose the learning problem
- Agent: Use multi-agent ensemble to further improve

Questions for Discussion

- In your research field, what would be a good way to “specify” a task?
 - GPT-2 uses text: “translate from English to French”
 - GPT-3 uses examples
- In your research field, what is a good way to define “basic” elements?
 - ViT uses pixel patches
 - PointNet uses point clouds
 - GPT-3 uses byte-level BPE tokens
- In your research field, is there any natural supervision for tasks?